Kartik

Third Year Undergraduate
Department of Computer science and Engineering
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ACADEMIC QUALIFICATIONS

Year	Degree/Certificate	Institute	CPI/%
2022 - Present	B. Tech.	Indian Institute of Technology Kanpur	5.2
2021	Class XII (CBSE)	Jawahar Navodaya Vidhyalaya, Mahoba	91%
2019	Class X (CBSE)	Jawahar Navodaya Vidhyalaya, Mahoba	84.6%

SCHOLASTIC ACHIEVEMENTS

- Secured **All India Rank 3777** in **JEE Advanced 2022** among 1,60,000 shortlisted candidates *(2022)*
- Secured **All India Rank 7117** in JEE Main 2022 among 1.5 million applicants, conducted by NTA *(2022)*
- Secured **100 Percentile** in Chemistry JEE Main 2022 among 1.5 million applicants, conducted by NTA *(2022)*

KEY PROJECTS

Profinfo Central | Course Project| profinfo central (*Jan'24-Apr'24*)
Prof. Indranil saha | CS253 - Software Development and Operations
- Developed a platform that showcases projects offered by professors, simplifying the process for students to apply for projects.
- **Led** a team of 10 to develop a site where students can request projects, and professors can accept or reject project requests.
- Utilized **Express.js** for the backend , MongoDB for data handling, and **JWT** tokens for authentication.
- Implemented the front-end using **React.js** and its hooks for state management, with **Postman** used for API testing.

Assignment-1 | Course Project| mini-project-1 (*Aug'24-Sep'24*)
Prof. Piyush Rai | CS771 - Introduction to Machine Learning
- Built a machine learning pipeline for Emoticon Classification using Logistic Regression, LSTM, and RandomForest.
- Preprocessed data with emoji removal, tokenization, padding, and label encoding for sequence uniformity.
- Achieved 97% accuracy with RandomForest and 96.84% accuracy using RNN-LSTM-based feature extraction.
- Designed an LSTM sequence classifier with embedding and dense layers, achieving 86.92% accuracy.

Assignment-2 | Course Project| mini-project-2 (*Oct'24-Nov'24*)
Prof. Piyush Rai | CS771 - Introduction to Machine Learning
- Developed a binary classification model utilizing Decision Tree, XGBoost, and RNN for sequence-based data.
- Implemented advanced data preprocessing techniques, sequence encoding and feature engineering, to enhance performance.
- Achieved high accuracy using LSTM, surpassing 96%, optimizing hyperparameters through rigorous experimentation.
- Evaluated models with precision-recall metrics and visualized results for comprehensive performance analysis.

Anonymous chat-app | Self Project| Chat-App (*Nov'23-Dec'23*)
- Developed an anonymous chat web app using **React.js** ,**Node.js** ,**Express.js** , and Socket.io, enabling IITK students.
- Integrated **MongoDB** for efficient data storage, supporting student searches and filtering in the anonymous chat system.
- Designed gender-based filters enabling selective identity reveal, providing users control over their privacy in chats.
- Implemented **Socket.io** for real-time messaging, ensuring low-latency, scalable communication between IITK students.
- Secured access to IITK users only, implementing authentication for exclusive, safe interaction within the student community.

Forecasting using Time-Series | Stamatics, IIT Kanpur | Time-series (*May'24-Jul'24*)
- Utilized **ARIMA** models to predict Apple (AAPL) stock closing prices, with data split into 80:20 for training and testing.
- Applied **Probabilistic** methods to calculate stock price movements, market trends, and returns using time series data.
- Conducted **statistical hypothesis** testing to verify stock return and constructed intervals for precise financial forecasting.
- Decomposed **time series** data exhibiting trends and seasonality into additive components for better prediction accuracy.

TECHNICAL SKILLS

- **Programming Languages:** C, C++, javaScript, Python
- **Frameworks and Utilities:** Git, Github, React.js, Socket.io, LaTeX, HTML, CSS, jQuery, Node.js, Express.js, Postman, EJS, Bash, Numpy, Pandas, Matplotlib, Scikit-learn, Scipy, Tensor-flow, Pytorch, Keras, Seaborn

RELEVANT COURSES (**=Online Courses*)

Data Structures and Algorithms	Introduction to Machine Learning	Software Development and Operations
Theory of Computation	Fundamental of Computing	Computer Organisation
Introduction to Probability	Introduction to Electronics	Web development*(Udemy)

EXTRA-CURRICULAR ACTIVITIES

- Represented hall-12 with 5 team members and won in **Snt Code** in single hand **Speedcubing** competition